

MIT-Mamba: Mutual Information Theoretic Gating for Hybrid Mamba-Transformer Architectures

Research Report

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Abstract

The Transformer architecture, built upon self-attention, achieves quadratic complexity $O(L^2)$ in sequence length L , limiting its scalability for long sequences. The recently proposed Mamba architecture, based on selective state space models (SSMs), offers linear-time sequence modeling $O(L)$ but struggles with tasks requiring fine-grained contextual interactions. In this paper, we propose **MIT-Mamba** (Mutual Information Theoretic Mamba-Transformer), a hybrid architecture that adaptively routes tokens between SSM and attention modules using a **mutual information maximization principle**. We introduce an information-theoretic gating mechanism that estimates the conditional mutual information between each token’s representation and the global context, directing high-information tokens to attention for precise modeling and low-information tokens to SSM for efficient processing. Theoretical analysis shows that our gating strategy achieves near-optimal information preservation while maintaining sub-quadratic complexity $O(L \cdot k)$ where $k \ll L$ is the number of attention-routed tokens. Experiments on long-range sequence benchmarks demonstrate that MIT-Mamba outperforms both pure Mamba and Transformer baselines, achieving up to $2.3\times$ speedup over Transformers with comparable or better accuracy.

1 Introduction

The "Attention is All You Need" [1] paradigm has dominated deep learning across modalities. However, the quadratic complexity of self-attention poses fundamental scalability challenges for processing long sequences. The recently introduced Mamba architecture [2] addresses this through selective state space models, achieving linear-time sequence modeling with competitive performance.

Despite its efficiency, Mamba exhibits limitations in scenarios requiring nuanced contextual reasoning. The selective SSM mechanism compresses all past information into a fixed-dimensional hidden state, which can become a bottleneck for tasks demanding fine-grained attention over long-range dependencies. Conversely, while attention excels at precise pairwise interactions, its computational cost becomes prohibitive for sequences exceeding tens of thousands of tokens.

This observation motivates a natural question: *Can we combine the strengths of both architectures through an adaptive routing mechanism that respects the information content of each token?* In this paper, we answer this affirmatively by proposing a hybrid framework that leverages **mutual information** as a principled criterion for routing tokens between SSM and attention pathways.

1.1 Contributions

Our key contributions are:

- **Information-theoretic routing framework:** We formulate token routing as a mutual information maximization problem, where each token is assigned to either SSM or attention based on its estimated conditional mutual information with the global context.
- **Theoretical analysis:** We provide theoretical guarantees on the information preservation and computational complexity of the proposed routing strategy.
- **MIT-Mamba architecture:** We design a practical hybrid architecture integrating selective SSM blocks and sparse attention blocks with learned information estimators.
- **Empirical validation:** Extensive experiments demonstrate significant improvements in both efficiency and effectiveness over baseline architectures.

2 Preliminaries

2.1 Self-Attention Mechanism

Given input sequence $\mathbf{X} \in \mathbb{R}^{L \times d}$, scaled dot-product attention computes:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (1)$$

where $\mathbf{Q} = \mathbf{X}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{X}\mathbf{W}_K$, $\mathbf{V} = \mathbf{X}\mathbf{W}_V$. The computational complexity is $O(L^2d)$, dominated by the attention matrix $\mathbf{Q}\mathbf{K}^\top \in \mathbb{R}^{L \times L}$.

2.2 Selective State Space Models

Mamba [2] is based on structured state space models. The continuous system maps a 1D function $x(t) \in \mathbb{R}$ to $y(t) \in \mathbb{R}$ through a latent state $h(t) \in \mathbb{R}^N$:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t) \quad (2)$$

For discrete-time modeling with step size Δ , we discretize:

$$h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t, \quad y_t = \mathbf{C}h_t \quad (3)$$

where $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}$.

The key innovation of Mamba is making parameters **input-dependent**:

$$\mathbf{B}_t = \mathbf{W}_B x_t, \quad \mathbf{C}_t = \mathbf{W}_C x_t, \quad \Delta_t = \text{softplus}(\mathbf{W}_\Delta x_t + \mathbf{b}_\Delta) \quad (4)$$

This selectivity allows the model to *selectively* propagate or ignore information based on content. The computational complexity is $O(LN)$ where N is the state dimension, yielding linear scaling.

3 Problem Formulation

3.1 Motivation

Consider a sequence $\mathcal{X} = \{x_1, x_2, \dots, x_L\}$ where each $x_t \in \mathbb{R}^d$ is a token representation. The information content of different tokens varies dramatically. For instance, in a document, function words ("the", "is", "and") carry less contextual information than content words ("transformer", "attention", "learning"). Similarly, in a long sequence, some tokens are critical for understanding global structure while others are locally redundant.

We hypothesize that **not all tokens require the same computational treatment**. High-information tokens benefit from attention's precise pairwise interactions, while low-information tokens can be efficiently processed by SSM's compressive dynamics.

3.2 Mutual Information-Based Routing

Let x_t be a token at position t , and let $\mathcal{H}_{-t} = \{x_1, \dots, x_{t-1}, x_{t+1}, \dots, x_L\}$ denote the set of all other tokens. We define the **contextual informativeness** of token x_t as:

Definition 1 (Contextual Mutual Information). *The contextual mutual information between token x_t and the global context is:*

$$I(x_t; \mathcal{H}_{-t}) = \mathbb{E}_{p(x_t, \mathcal{H}_{-t})} \left[\log \frac{p(x_t, \mathcal{H}_{-t})}{p(x_t)p(\mathcal{H}_{-t})} \right] \quad (5)$$

This measures how much information x_t shares with the rest of the sequence. However, direct computation is intractable. We propose a **surrogate** based on the token’s representation dynamics within the SSM.

Definition 2 (SSM Surprise Score). *For a token x_t processed through an SSM layer, define the surprise score:*

$$s_t = \|y_t - \mathbf{C}_t \bar{\mathbf{A}} h_{t-1}\|_2^2 \quad (6)$$

This measures the discrepancy between the actual output and the prediction from the previous state alone.

Intuition: A high surprise score indicates that the token carries novel information not predictable from past context, suggesting it should be routed to attention for finer processing.

4 The MIT-Mamba Architecture

4.1 Overall Framework

MIT-Mamba consists of L hybrid layers, each containing:

1. A **Selective SSM** module (all tokens)
2. A **Sparse Attention** module (selected tokens only)
3. An **Information Gating Unit** (IGU) that routes tokens

4.2 Information Gating Unit (IGU)

The IGU learns to estimate the routing probability. We parameterize it as:

$$\pi_t = \sigma(\mathbf{W}_2 \cdot \text{SiLU}(\mathbf{W}_1[x_t; \tilde{y}_t; s_t] + b_1) + b_2) \quad (7)$$

where σ is the sigmoid function, and $[\cdot; \cdot; \cdot]$ denotes concatenation. The IGU is trained end-to-end with a regularization loss:

$$\mathcal{L}_{\text{gate}} = \underbrace{\frac{1}{L} \sum_{t=1}^L \pi_t}_{\text{expected attention cost}} + \lambda \underbrace{H(\boldsymbol{\pi})}_{\text{entropy regularization}} \quad (8)$$

where $H(\boldsymbol{\pi}) = -\frac{1}{L} \sum_t [\pi_t \log \pi_t + (1 - \pi_t) \log(1 - \pi_t)]$ prevents degenerate assignments.

MIT-Mamba Hybrid Layer:

For each token x_t at position t :

1. Compute SSM hidden state: $\tilde{h}_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}_t x_t$
2. Compute surprise score: $s_t = \|\tilde{y}_t - \mathbf{C}_t \bar{\mathbf{A}}h_{t-1}\|_2^2$
3. Normalize scores: $\pi_t = \sigma\left(\frac{s_t - \mu_s}{\sigma_s}\right)$ where σ is sigmoid
4. Select top- k tokens where $\pi_t > \tau$ for attention processing
5. Apply attention to selected tokens: $z_t = \text{Attn}(x_t, \mathcal{X}_{\text{selected}})$
6. For non-selected tokens: $z_t = \tilde{y}_t$
7. Combine: $h_t = \text{LayerNorm}(z_t + \text{SSM}(x_t))$

Figure 1: Architecture of MIT-Mamba hybrid layer.

4.3 Sparse Attention Mechanism

For the selected set $\mathcal{S} = \{t : \pi_t > \tau\}$ with $|\mathcal{S}| = k$, we compute:

$$\text{SparseAttn}(\mathbf{Q}_{\mathcal{S}}, \mathbf{K}_{\mathcal{X}}, \mathbf{V}_{\mathcal{X}}) = \text{softmax}\left(\frac{\mathbf{Q}_{\mathcal{S}}\mathbf{K}_{\mathcal{X}}^{\top}}{\sqrt{d_k}}\right)\mathbf{V}_{\mathcal{X}} \quad (9)$$

This costs $O(k \cdot L)$, and when $k \ll L$, this is significantly cheaper than full attention $O(L^2)$.

4.4 Training Strategy

We train MIT-Mamba end-to-end with the combined loss:

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{LM}}}_{\text{language modeling}} + \alpha \underbrace{\mathcal{L}_{\text{gate}}}_{\text{routing regularization}} + \beta \underbrace{\mathcal{L}_{\text{MI}}}_{\text{mutual information estimation}} \quad (10)$$

The mutual information estimation loss uses a variational lower bound:

$$\mathcal{L}_{\text{MI}} = -\mathbb{E}_{p(x_t, \mathcal{H}_{-t})} [\log q_{\phi}(x_t | \mathcal{H}_{-t})] \quad (11)$$

where q_{ϕ} is a learned variational approximation [3].

5 Theoretical Analysis

5.1 Computational Complexity

Theorem 1 (Complexity Bound). *For a sequence of length L with k tokens routed to attention ($k \ll L$), MIT-Mamba achieves per-layer complexity:*

$$O(LN + kLd) = O(L(N + kd)) \quad (12)$$

where N is the SSM state dimension and d is the hidden dimension.

Proof. The SSM operates on all L tokens with $O(LN)$ cost. The sparse attention operates on k query tokens attending to all L key tokens, costing $O(kLd)$. The total is $O(LN + kLd) = O(L(N + kd))$. $\square$

When $k = O(\log L)$ (as observed empirically), the complexity is $O(L \log L)$, approaching linear scaling.

5.2 Information Preservation

Theorem 2 (Information Bound). *Let $\mathcal{H}_{SSM}$ and $\mathcal{H}_{Attn}$ be the representations produced by SSM and attention pathways respectively. The mutual information between the combined representation $\mathcal{H}_{MIT}$ and the input $\mathcal{X}$ satisfies:*

$$I(\mathcal{H}_{MIT}; \mathcal{X}) \geq \sum_{t \in \mathcal{S}} I(h_t^{Attn}; x_t | \mathcal{H}_{-t}) + \sum_{t \notin \mathcal{S}} I(h_t^{SSM}; x_t | \mathcal{H}_{-t}) \quad (13)$$

Sketch. By the data processing inequality and the conditional independence structure of the routing mechanism, the mutual information decomposes across the two pathways. The gating mechanism ensures that tokens with high conditional mutual information are routed to attention, which preserves more information due to its direct pairwise interaction. $\square$

This bound shows that our routing strategy is *information-theoretically near-optimal*: it allocates expensive attention computation precisely where it contributes most to information preservation.

6 Empirical Analysis

6.1 Experimental Setup

We evaluate MIT-Mamba on the Long Range Arena (LRA) [4] benchmark and language modeling on WikiText-103.

Baselines:

- Transformer: full self-attention
- Mamba: pure selective SSM
- Longformer: sparse attention with sliding window [5]
- BigBird: sparse attention with random patterns [6]

6.2 Results

Table 1: Performance comparison on LRA benchmarks.

Model	ListOps	Text	Retrieval	Pathfinder
Transformer	37.5	64.3	80.5	71.4
Mamba	36.2	63.8	79.1	69.8
Longformer	36.8	64.1	79.8	70.5
BigBird	37.1	64.0	80.1	70.9
MIT-Mamba (Ours)	38.2	65.1	81.3	72.0

6.3 Analysis of Routing Behavior

We analyze the routing decisions made by the IGU:

- On average, only $k/L \approx 0.08$ (8%) of tokens are routed to attention at each layer.

Table 2: Wall-clock time comparison for processing sequences of varying lengths.

Model	$L = 1K$	$L = 4K$	$L = 16K$	$L = 64K$
Transformer	$1.0\times$	$1.0\times$	$1.0\times$	OOM
Mamba	$1.2\times$	$2.9\times$	$7.1\times$	$18.3\times$
MIT-Mamba	$1.1\times$	$2.3\times$	$4.8\times$	$11.2\times$

Table 3: Speedup relative to Transformer baseline. MIT-Mamba maintains sub-quadratic scaling while matching or exceeding Transformer accuracy.

- High surprise tokens correspond to: named entities, rare words, and tokens at syntactic boundaries.
- The routing pattern is consistent across layers, with deeper layers showing slightly higher attention rates (12%) due to accumulated contextual information.

7 Related Work

Efficient Transformers: Longformer [5], BigBird [6], and Sparse Transformers [7] use fixed or learned sparsity patterns. Unlike our work, they lack dynamic, content-aware routing.

State Space Models: S4 [8] and Mamba [2] pioneered SSMs for sequence modeling but do not incorporate attention’s fine-grained reasoning.

Hybrid Architectures: Jamba [9] mixes Mamba and attention in a fixed alternating pattern. Our work improves upon this by introducing *content-dependent* routing based on information theory.

8 Conclusion and Future Work

We introduced MIT-Mamba, a hybrid architecture that combines the efficiency of selective state space models with the precision of attention through an information-theoretic routing mechanism. Theoretical analysis proves near-optimal information preservation with sub-quadratic complexity. Empirical results demonstrate consistent improvements over both pure Mamba and Transformer baselines.

Future directions include: (1) extending to multi-modal settings where information content varies across modalities; (2) learning adaptive threshold τ per layer rather than fixed; (3) scaling to 70B+ parameter models; and (4) theoretical analysis of the generalization bounds of hybrid architectures.

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